Scale Without Law: Why the de Sitter–Unruh Temperature Forces the MOND Acceleration but Not the Interpolation

Carl P. Zimmerman · Briar Creek Tech · 2026-06-28

Abstract

The MOND acceleration scale satisfies $a_0 \approx c^2\sqrt{(\Lambda/32\pi)} = cH\_\Lambda/Z$ to within a factor of order unity — the long-noted coincidence that the scale of galactic dynamics equals a scale built from the cosmological constant. Milgrom (1999) proposed reading this as a vacuum effect: an observer accelerating in de Sitter space sees the Deser–Levin temperature $T(a) \propto \sqrt{(a^2 + a\_\Lambda^2)}$, with $a\_\Lambda = cH\_\Lambda$, and the *excess* over the de Sitter background, $\Delta T = \sqrt{(a^2+a\_\Lambda^2)} - a\_\Lambda$, has the asymptotic form of a MOND inertia. This note asks, with explicit and adversarially-checked calculation, **how far that mechanism actually goes** — and reports an honest, two-sided result. The de Sitter–Unruh temperature **forces the scale** ($a_0 \sim c\sqrt{\Lambda}$ is recovered, not assumed) and yields the **deep-MOND √-law and the baryonic Tully–Fisher relation** (both exact). It does **not** yield the full interpolation function, the precise coefficient, or a covariant equation of motion — the standing open problem since Milgrom 1999. We make the obstructions precise and contribute three results: (i) the candidate interpolation $\mu\_fw = (\sqrt{(1+4x^2)}-1)/2x$ and the bare dS–Unruh interpolation $\mu = (\sqrt{(1+x^2)}-1)/x$ are the **same one-parameter family**, separated by a single rescaling, so the whole gap collapses to one undetermined coefficient Z; (ii) a **number-field obstruction** — thermodynamic normalizations lie in $\mathbb{Q}(\pi)$ while $Z = \sqrt{(32\pi/3)}$ lies in $\mathbb{Q}(\sqrt{\pi})$ — proving that no horizon-entropy or temperature factor can ever output the coefficient; (iii) a **strengthened sign theorem** showing the MOND-signed response requires breaking ghost-freedom or supplying an active drive, which blocks a passive-bath covariant completion. The conclusion is deliberately modest: the de Sitter–Unruh route owns the *scale* and the deep-MOND *limit* but not the *law*, and the residual freedom is now reduced to a single, provably-unforced number. This is offered as a map of the obstruction, not a derivation of MOND, and not a theory of anything beyond gravity.

1. What the mechanism delivers

Take the Deser–Levin temperature for an observer of proper acceleration a in de Sitter space (Deser & Levin 1997; Narnhofer, Peter & Thirring 1996),

$$T(a) = (\hbar/2\pi c k\_B) \sqrt{(a^2 + a\_\Lambda^2)}, \quad a\_\Lambda = cH\_\Lambda,$$

and Milgrom's (1999) postulate that inertia responds to the excess over the de Sitter vacuum, $\Delta T(a) = \sqrt{(a^2+a\_\Lambda^2)} - a\_\Lambda$. Two things follow without further input (`reviews/deser_levin_mond_derivation.py`, exit 0):

- **The scale is forced.** The transition between the two regimes occurs at $a \sim a\_\Lambda = cH\_\Lambda$; the acceleration scale is *built from Λ*, not inserted. The $a_0 \sim c\sqrt{\Lambda}$ coincidence becomes a consequence.

- **The deep-MOND limit is exact.** For $a \ll a\_\Lambda$, $\Delta T \approx a^2/2a\_\Lambda$; with the inertial force $\propto \Delta T$, requiring it equal the Newtonian g_bar gives $a = \sqrt{(2\ a\_\Lambda\ g\_bar)}$ — the deep-MOND square-root law. The baryonic Tully–Fisher relation $V^4 = G\ M\ a_0$ follows (symbolically exact).

This is real, and it is the genuine content of Milgrom's proposal. It is also where the easy progress ends.

2. The interpolation: one family, one coefficient (Gap A)

The framework's interpolation, written as $g\_obs = \sqrt{(g\_bar^2 + g\_bar\ a_0)}$, is equivalent to the convex function $\mu\_fw(x) = (\sqrt{(1+4x^2)}-1)/(2x)$ with $x = a/a_0$. The bare excess-temperature postulate of §1, by contrast, yields $\mu(x) = (\sqrt{(1+x^2)}-1)/x$. These look different, and a referee would ask which the mechanism actually predicts. The answer (`reviews/close_mu_from_temperature.py`, exit 0, sympy):

> **They are the same one-parameter family.** $\mu\_dl(x) = \mu\_fw(x/2)$ exactly; the only difference is a rescaling of the acceleration unit, i.e. a redefinition of the coefficient relating a_0 to a_Λ. The "two interpolation gaps" are one gap: the undetermined number Z.

We tested six principled couplings of inertia to T(a). One — the free-energy reading $\varphi = -T(a)$, $\mu = d\varphi/da = x/\sqrt{(1+x^2)}$ — is genuinely principled, has the correct Newtonian and deep-MOND limits, *and* the correct (MOND) sign. But it produces the **simple-ν** interpolation, not μ_fw, and the choice of free energy is not forced. Reaching μ_fw *exactly* requires inserting the coefficient by hand. That this is reverse-engineering and not derivation we prove constructively: the floor-fit that "recovers" the framework's form reproduces an entire decoy family $g\_obs^2 = g\_bar^2 + k \cdot g\_bar \cdot a_0$ for $k = ½, 1, 3, 7/4$ *equally well* — it has zero power to discriminate the coefficient. **The interpolation is a posit; what the mechanism fixes is the family, not the member.**

3. The coefficient: a number-field obstruction (Gap B)

The bare mechanism's coefficient comes out $a_0\_eff = 2a\_\Lambda = 2cH\_\Lambda$; the framework's is $a_0 = cH\_\Lambda/Z$ with $Z = \sqrt{(32\pi/3)} \approx 5.789$. Can a defensible normalization — the Unruh 2π, the Bekenstein–Hawking ¼, a surface-gravity ½, the dimension factor $d(d-1) = 6$ — bridge the factor 2Z? We checked each (`reviews/close_coefficient_Z.py`, exit 0). None does, and there is a clean reason:

> **Thermodynamic normalizations live in the field $\mathbb{Q}(\pi)$; $Z = \sqrt{(32\pi/3)}$ lives in $\mathbb{Q}(\sqrt{\pi})$. The two fields do not meet.** A factor assembled from 2π, ¼, 4π, 6, … is rational-in-π and can *bracket* Z (indeed 6 and 2π straddle 5.79) but can never *equal* it. Only an intrinsic $\sqrt{\rho}$ route — $a_0 = (c/2)\sqrt{(G\rho\_DE)}$ — lands on Z exactly, and there the ½ prefactor and the ρ_DE-vs-ρ_total choice are provably free ($Z \propto 1/\kappa$).

The tempting factorization $Z^2 = 4\cdot(8\pi/3)$ does not rescue it: carried symbolically, the "4" is $1/\kappa^2$ and equals 4 only at the posited $\kappa = ½$ — it is the free prefactor squared, not the area-law ¼. **The coefficient is defined, not derived. It is the one number everything reduces to, and it is unforced.**

4. The equation of motion: a negative theorem (Gap C)

A genuine modified-inertia theory needs an action giving $F = m\ \mu(a/a_0)\ a$. We re-examined the three routes adversarially (`reviews/close_covariant_eom.py`, `verify_closure_covariant_eom_adversarial.py`, both exit 0, 11/11 attacks survived):

- **Local, time-nonlocal action** → Ostrogradsky ghost (the worldline Hessian is non-degenerate; the DHOST/auxiliary-field escape Legendre-transforms the nonlinearity straight back unless the coupling itself carries the time-nonlocality, i.e. it *is* the nonlocal route).
- **Field-theoretic completion** (khronon/Galileon) → the static slip carries a non-vanishing divergence, fixed by the Solar-System (Cassini) bound; a divergence-free completion is radiative and produces no static MOND force.
- **Nonlocal worldline action** → ghost-free, but the MOND-signed kernel must be **active**: a passive de Sitter bath gives a positive mass renormalization, $\delta m = 2\int \rho(\omega)/\omega^2\, d\omega \geq 0$, i.e. the *anti-MOND* sign. We strengthen the standing result: this needs **only ghost-freedom of the bath modes**, not passivity or fluctuation–dissipation; the finite-frequency apparent-escape lives in the Newtonian band and falls as $1/\omega^2$, the wrong monotonicity for MOND.

There is no fourth route, and the no-go uses none of {Z, a_0, μ_fw} — it would even forbid an anti-MOND theory. The single honest crack is external: a **non-de-Sitter, in-band, phase-coherent drive** (a “galactic pump”) could supply the active kernel — but none is known, and it would still owe an explanation of a_0′s universality.

5. The one observational handle

The mechanism’s distinctive, falsifiable content is not the static algebra but the *time* dependence. If a_0 is genuinely set by the dark-energy density, then $a_0(z) \propto \sqrt{\rho_{DE}(z)}$: on the DESI dynamical-dark-energy branch it declines (~25–40 % by z = 3); under a true cosmological constant ($w = -1$) it is constant. This is the one place the de Sitter–Unruh reading is separable from a static coincidence — and it dies cleanly if dark energy proves to be Λ.

6. Conclusion

The de Sitter–Unruh temperature forces the MOND **scale** and reproduces the deep-MOND **limit** — Milgrom’s 1999 insight, made exact. It does not produce the **interpolation** (which collapses to a single undetermined coefficient), the **coefficient** itself (number-field-obstructed; defined, not derived), or a **covariant equation of motion** (blocked by an Ostrogradsky/Cassini/sign-theorem trichotomy). The honest standing is therefore unchanged from 1999 in substance, but sharpened in form: the gap between “the scale” and “the law” is now reduced to one provably-unforced number and a clean sign obstruction, both stated as theorems rather than impressions. We claim no derivation of MOND, no fixing of the coefficient, and nothing about the Standard Model; the value here is a precise map of where an inertia-from-vacuum programme must still do work, and the identification of the single quantity on which everything turns.

References (representative, real)

- M. Milgrom, *The modified dynamics as a vacuum effect*, Phys. Lett. A 253 (1999) 273.

- S. Deser, O. Levin, *Accelerated detectors and temperature in (anti−)de Sitter spaces*, Class. Quantum Grav. 14 (1997) L163.
- H. Narnhofer, I. Peter, W. Thirring, *Unruh effect in de Sitter space*, Int. J. Mod. Phys. B 10 (1996) 1507.
- B. Famaey, S. McGaugh, *Modified Newtonian Dynamics: Observational Phenomenology and Relativistic Extensions*, Living Rev. Relativity 15 (2012) 10.
- M. Milgrom, *MOND theory*, Can. J. Phys. 93 (2015) 107 (modified-inertia / nonlocal formulations).

Reproducible: `reviews/{deser_levin_mond_derivation, close_mu_from_temperature, close_coefficient_Z, close_covariant_eom, verify_closure_gapA_adversarial, verify_closure_covariant_eom_adversarial}.py` *(all exit 0). Footing* $a_0 = c^2\sqrt{(\Lambda/32\pi)} = cH\_\Lambda/Z$, $Z = \sqrt{(32\pi/3)}$. *The author retracted all earlier "theory of everything" claims (2026-06-23); this note makes none, and is a foundations/obstruction map, not a derivation.*